

HR EMPLOYEE ANALYTICS

Data Analytics Project Report

Prepared By

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Technology Stack

Python • Excel • MySQL • Power BI • Power Query • DAX

Project Dataset: 1,500 Employee Records

An end-to-end HR analytics project for workforce, salary, performance, satisfaction, and attrition analysis.

1. Executive Summary

The **HR Employee Analytics** project is an end-to-end data analytics solution designed to transform employee-level HR data into structured, interactive and business-oriented insights. The project combines Python for data preparation, Excel for structured data storage and documentation, MySQL for relational database management, and Power BI for interactive reporting and visualization.

The analytical dataset contains **1,500 employee records**. The project examines workforce size, active employees, employee attrition, salary, age, tenure, job roles, departments, overtime, performance, job satisfaction, work-life balance, and training. The final Power BI solution contains multiple report pages, KPI cards, slicers, drill-through functionality, tooltips, bookmarks, and an HR Insights page.

The analysis supports evidence-based HR review. Reported patterns represent relationships in the project dataset and should not automatically be interpreted as causal effects.

Key Project Metrics

Metric	Value	Meaning
Total Employees	1,500	Total employee records
Active Employees	1,284	Employees with Attrition = No
Attrition Count	216	Employees with Attrition = Yes
Attrition Rate	14.40%	Attrition Count ÷ Total Employees
Average Age	33.77 years	Average employee age
Average Tenure	4.11 years	Average YearsAtCompany
Average Monthly Salary	≈ 56.59K	Average MonthlyIncome

2. Project Objectives

- Create a realistic employee dataset suitable for HR analytics and portfolio demonstration.
- Clean, transform and enrich employee data using Python.
- Store and organize data in a relational MySQL database.
- Create department and job-role lookup structures and a final analytical employee table.
- Develop reusable SQL queries for workforce, salary, attrition, performance and satisfaction analysis.
- Build a Power BI analytical model using fact and dimension tables.
- Create DAX measures for major HR KPIs.
- Develop interactive dashboards with slicers, drill-through, tooltips, bookmarks and navigation.
- Document the dataset, business rules, database architecture and analytical workflow.

3. Project Scope

Area	Covered in Project
Workforce	Employee count, age, gender, department, job role, education and job level
Attrition	Attrition count, attrition rate and segmentation by HR dimensions
Salary	Monthly income, salary bands and salary comparisons

Area	Covered in Project
Tenure	Years at company, tenure bands and tenure-based attrition analysis
Performance	Performance ratings and high-performance employee analysis
Satisfaction	Job satisfaction, environment satisfaction and work-life balance
Training	Training hours and employee training analysis
Work Pattern	Overtime and business travel analysis
Reporting	Interactive Power BI dashboards and HR Insights

4. Technologies and Tools

Technology / Tool	Role in Project
Python	Dataset generation, preprocessing, transformation and MySQL import automation
Pandas	Tabular data processing and validation
Excel	Workbook-based data storage, review and data dictionary
MySQL	Relational database, staging, dimensions, final employee table and SQL analysis
MySQL Workbench	Database creation, SQL execution and validation
Power Query	Data preparation and model-loading workflow in Power BI
Power BI	Interactive dashboard, data model and visualization
DAX	HR KPI measures and analytical calculations

5. End-to-End Architecture

Source → Python → Excel/CSV → MySQL Staging → Dimension Mapping → Final Employee Table → Power BI Model → DAX → Dashboard → HR Insights

Layer	Object / Process	Purpose
1	HR Employee Data.xlsx / CSV	Source employee records
2	Python preprocessing	Create and validate AgeGroup, SalaryBand and TenureBand
3	employees_staging	Temporary raw-data layer in MySQL
4	departments	Department lookup dimension
5	job_roles	Job-role lookup dimension
6	employees	Final analytical employee table with foreign keys
7	Power BI model	FactEmployee + DimDepartment + DimJobRole + DimDate
8	DAX	KPI calculations and analytical measures
9	Power BI reports	Interactive HR dashboards and insights

6. Dataset Description

The project uses a 1,500-record employee dataset designed for HR analytics. Each row represents an employee and contains demographic, organizational, compensation, satisfaction, performance, training, work-pattern, employment-date and attrition information.

Category	Representative Fields
Identity	EmployeeID, EmployeeName
Demographics	Gender, Age, MaritalStatus, DistanceFromHome
Organization	DepartmentID, JobRoleID, Education, JobLevel
Compensation	MonthlyIncome, SalaryBand
Experience	TotalWorkingYears, YearsAtCompany, NumCompaniesWorked
Career Progression	YearsInCurrentRole, YearsSincePromotion
Satisfaction	JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance
Performance	PerformanceRating, TrainingHours
Work Pattern	Overtime, BusinessTravel
Employment Lifecycle	HireDate, ExitDate, Attrition
Derived Segments	AgeGroup, SalaryBand, TenureBand

7. Python Data Preparation

Python is used as the data preparation layer. The workflow generates or loads the employee dataset, applies structured transformations, creates derived analytical categories, writes the data to Excel/CSV, and imports the staging dataset into MySQL.

Major preprocessing activities

- Generate or load employee records.
- Validate employee-level columns and data types.
- Create AgeGroup from Age.
- Create SalaryBand from MonthlyIncome using project-defined thresholds.
- Create TenureBand from YearsAtCompany.
- Preserve date fields as proper date values.
- Export clean data to CSV and Excel.
- Load the staging dataset into MySQL.

8. MySQL Database Design

MySQL provides the relational storage and SQL analysis layer. The project uses a staging table to preserve source records and separate lookup tables for departments and job roles. The final **employees** table stores department and job-role identifiers rather than repeating lookup text.

Table	Purpose	Important Relationships
employees_staging	Raw/staging employee data	Contains Department and JobRole text
departments	Department lookup	DepartmentID referenced by employees
job_roles	Job-role lookup	JobRoleID referenced by employees
employees	Final analytical employee table	FKs to departments and job_roles

Relational mapping

The staging Department value is matched to **departments.DepartmentName**, while the staging JobRole value is matched to **job_roles.JobRole**. The final employee table stores the corresponding DepartmentID and JobRoleID.

9. SQL Analysis

The SQL analysis layer is organized into reusable queries rather than retaining temporary verification scripts. The final project structure separates database setup, analytical queries and reusable views.

File	Purpose
database_setup.sql	Database/table creation and reproducible population workflow
data_analysis.sql	Business and HR analytical queries
views.sql	Reusable analytical views for reporting

Major SQL analyses

- Total employee and active employee counts.
- Attrition count and attrition rate.
- Employees and attrition by department.
- Employees and attrition by job role.
- Average salary by department and job role.
- Salary-band distribution and attrition.
- Age-group distribution and attrition.
- Tenure-band distribution and attrition.
- Overtime distribution and attrition.
- Gender distribution and attrition.
- Performance-rating analysis.
- Job-satisfaction and work-life-balance analysis.
- Training-hours analysis.

10. Power BI Data Model

The Power BI model follows a star-schema-style structure. The employee-level fact table is connected to descriptive dimensions for department, job role and date. This structure supports filtering, aggregation and reusable DAX measures.

Model Object	Role
FactEmployee	Employee-level fact/analytical table containing measurable HR attributes
DimDepartment	Department lookup and descriptive attributes
DimJobRole	Job-role lookup and descriptive attributes
DimDate	Date dimension for time-based reporting

11. DAX Measures

Core KPI definitions

Measure	Business Definition
Total Employees	COUNT of EmployeeID / employee rows
Active Employees	Employees where Attrition = No
Attrition Count	Employees where Attrition = Yes
Attrition Rate	Attrition Count ÷ Total Employees
Average Age	Average Age
Average Tenure	Average YearsAtCompany
Average Salary	Average MonthlyIncome

The measures are used across KPI cards and visuals so that filtering by department, job role, salary band, tenure, overtime and other dimensions dynamically updates the report.

12. Power BI Report Pages

Page	Main Purpose
Executive Overview	High-level workforce KPIs and management overview
Attrition Analysis	Attrition rate/count and segmentation across HR dimensions
Workforce & Salary	Workforce composition and salary distribution
Performance & Satisfaction	Performance ratings, satisfaction and training-related analysis
Employee Details	Detailed employee-level information with drill-through
HR Insights	Evidence-based patterns and business-oriented observations

Interactive Features

- Slicers for interactive filtering.
- Employee Details drill-through using EmployeeID.
- Back navigation from drill-through pages.
- Report-page tooltips for contextual detail.
- Bookmarks for filter-panel and reset-filter behavior.
- Page navigation for moving between report sections.

13. Key Dataset Findings

Indicator	Observed Value	Interpretation
Workforce size	1,500	The dataset contains 1,500 employee records.
Active workforce	1,284	Most records are marked Attrition = No.
Attrition	216	216 records are marked Attrition = Yes.
Overall attrition rate	14.40%	Calculated as $216 \div 1,500 \times 100$.
Average age	33.77 years	Average age in the dataset.
Average tenure	4.11 years	Average YearsAtCompany.
Average salary	≈ 56.59K	Average MonthlyIncome.

Department, job-role, salary-band, tenure-band, overtime and satisfaction patterns are explored in the HR Insights page. These comparisons identify areas for further HR investigation but do not by themselves establish why an employee leaves.

14. Data Dictionary Summary

The accompanying **Data Dictionary.xlsx** documents the analytical fields, source tables, data types, business meanings, transformations, relationships, business rules and data architecture.

Field Group	Examples
Identifiers	EmployeeID, DepartmentID, JobRoleID
Demographics	Gender, Age, MaritalStatus, DistanceFromHome
Organization	DepartmentID, JobRoleID, Education, JobLevel
Compensation	MonthlyIncome, SalaryBand
Tenure & Experience	TotalWorkingYears, YearsAtCompany, YearsInCurrentRole
Satisfaction	JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance
Performance	PerformanceRating, TrainingHours
Lifecycle	HireDate, ExitDate, Attrition
Derived Fields	AgeGroup, SalaryBand, TenureBand

15. Data Validation and Quality Checks

- Validated that the staging dataset contains 1,500 records.
- Validated that the final employees table contains 1,500 records.
- Validated department and job-role mappings before loading the final table.
- Used normalized matching with trimming and case-insensitive comparison during relational mapping.
- Validated final employee records through joins with department and job-role lookup tables.
- Checked KPI totals against the Power BI model.

16. Limitations

- The employee dataset is a project/portfolio dataset and should not be treated as confidential organizational HR data.
- Patterns in the dataset describe association, not causation.
- Salary values are analytical dataset values and should not be interpreted as current market compensation benchmarks.
- Attrition categories are based on the project dataset's Attrition field and definitions.
- HR decisions should use additional organizational context, qualitative feedback and validated operational data.

17. Future Enhancements

- Connect the dashboard to a live HR database or scheduled data pipeline.
- Add monthly/quarterly attrition trends using a full date dimension.
- Add employee cost and replacement-cost analysis.

- Add predictive attrition modeling with appropriate validation and governance.
- Add row-level security for HR/admin users.
- Automate dataset refresh and Power BI deployment.
- Add automated data-quality checks for duplicate IDs, missing fields and invalid dates.

18. Conclusion

The HR Employee Analytics project demonstrates a complete analytics workflow from data generation and preprocessing to relational database design, SQL analysis, Power BI modeling and interactive reporting. With 1,500 employee records, the solution provides a structured view of workforce composition, attrition, salary, tenure, performance and employee satisfaction.

The project also demonstrates practical skills in Python, Excel, SQL, MySQL, Power BI, Power Query and DAX. Its modular structure makes the work suitable for portfolio presentation while allowing future expansion into automated refreshes, advanced analytics and predictive HR use cases.

19. Recommended GitHub Project Structure

HR-Employee-Analytics/

■ ■ ■ ■ README.md

■ ■ ■ ■ Dataset/

■ ■ ■ ■ HR_Employee_Data.csv

■ ■ ■ ■ HR_Employee_Data.xlsx

■ ■ ■ ■ Python/

■ ■ ■ ■ generate_hr_dataset.py

■ ■ ■ ■ create_excel_workbook.py

■ ■ ■ ■ import_excel_to_mysql.py

■ ■ ■ ■ SQL/

■ ■ ■ ■ database_setup.sql

■ ■ ■ ■ data_analysis.sql

■ ■ ■ ■ views.sql

■ ■ ■ ■ PowerBI/

■ ■ ■ ■ HR_Employee_Analytics.pbix

■ ■ ■ ■ Screenshots/

■ ■ ■ ■ executive-overview.png

■ ■ ■ ■ attrition-analysis.png

■ ■ ■ ■ workforce-salary.png

■ ■ ■ ■ performance-satisfaction.png

■ ■ ■ ■ employee-details.png

■ ■ ■ ■ hr-insights.png

■ ■ ■ ■ Documentation/

■ ■ ■ ■ Data_Dictionary.xlsx

■ ■ ■ ■ Project_Report.pdf

■ ■ ■ ■ .gitignore

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